ELIZA · embedding experiment on the val split (KAN-97, KAN-221, KAN-99)

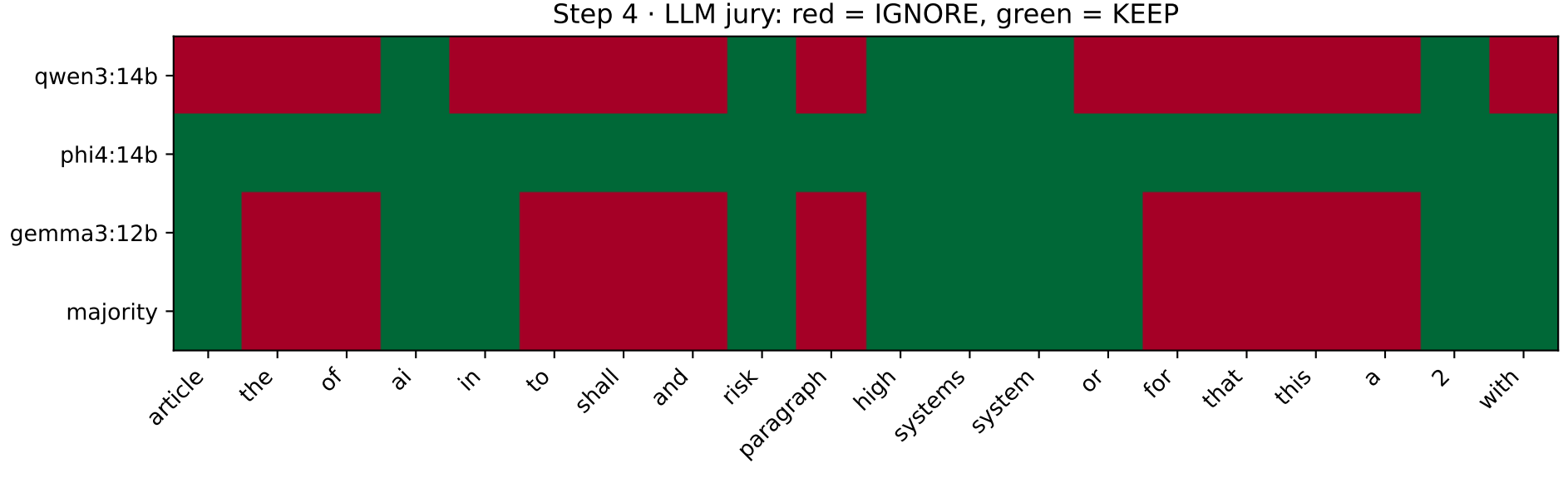
Val questions scored: 27

Embedding model: BAAI/bge-large-en-v1.5 dense weight: 0.9 BM25 words removed: stopwords

Words removed before embedding: none top-k: 8

Final search: Capped recall@5 0.51, Capped recall@8 0.58, MRR 0.69, Precision@5 0.24, Precision@8 0.19, Recall@5 0.48, Recall@8 0.56

W&B run: https://wandb.ai/gozdenur104-metu-middle-east-technical-university/eliza-rag/runs/oyrsrqwf


Step 4 · LLM jury: red = IGNORE, green = KEEP
qwen3:14b
phi4:14b
gemma3:12b
majority
article
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of
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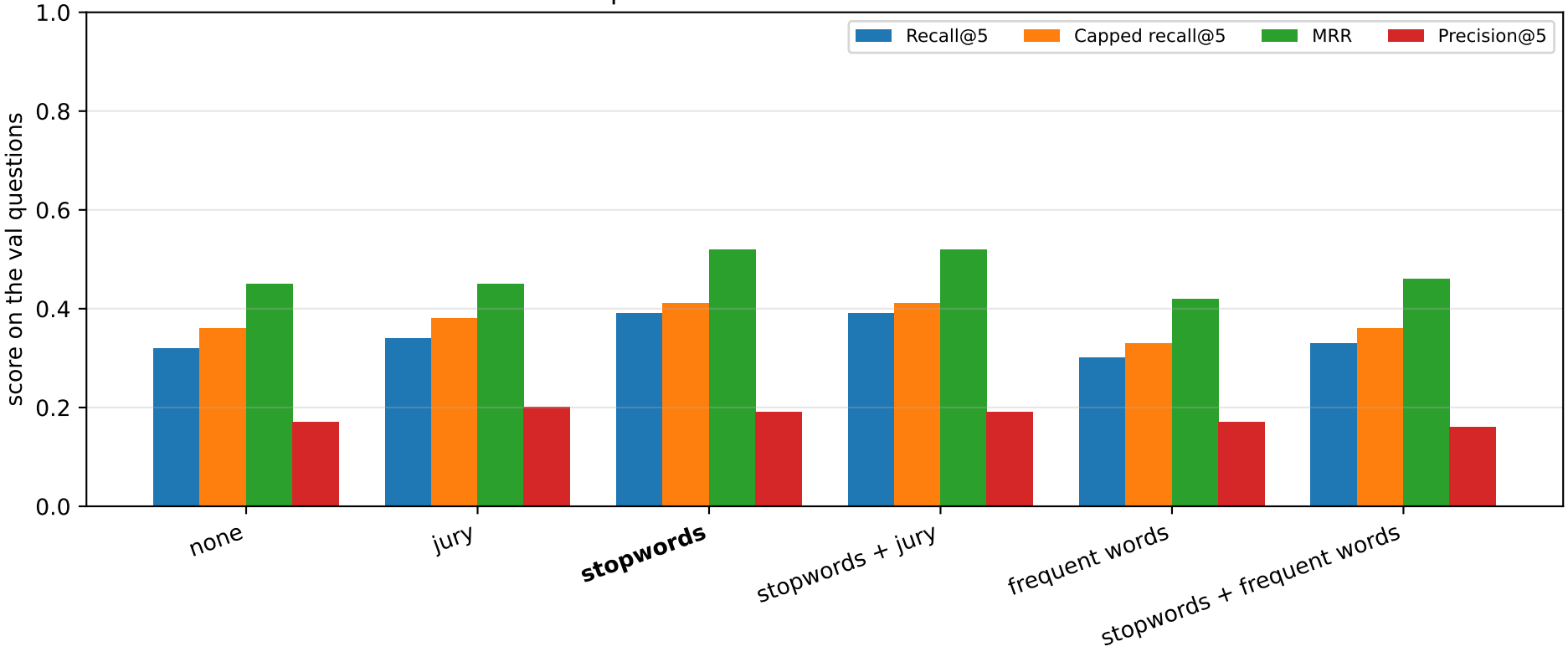
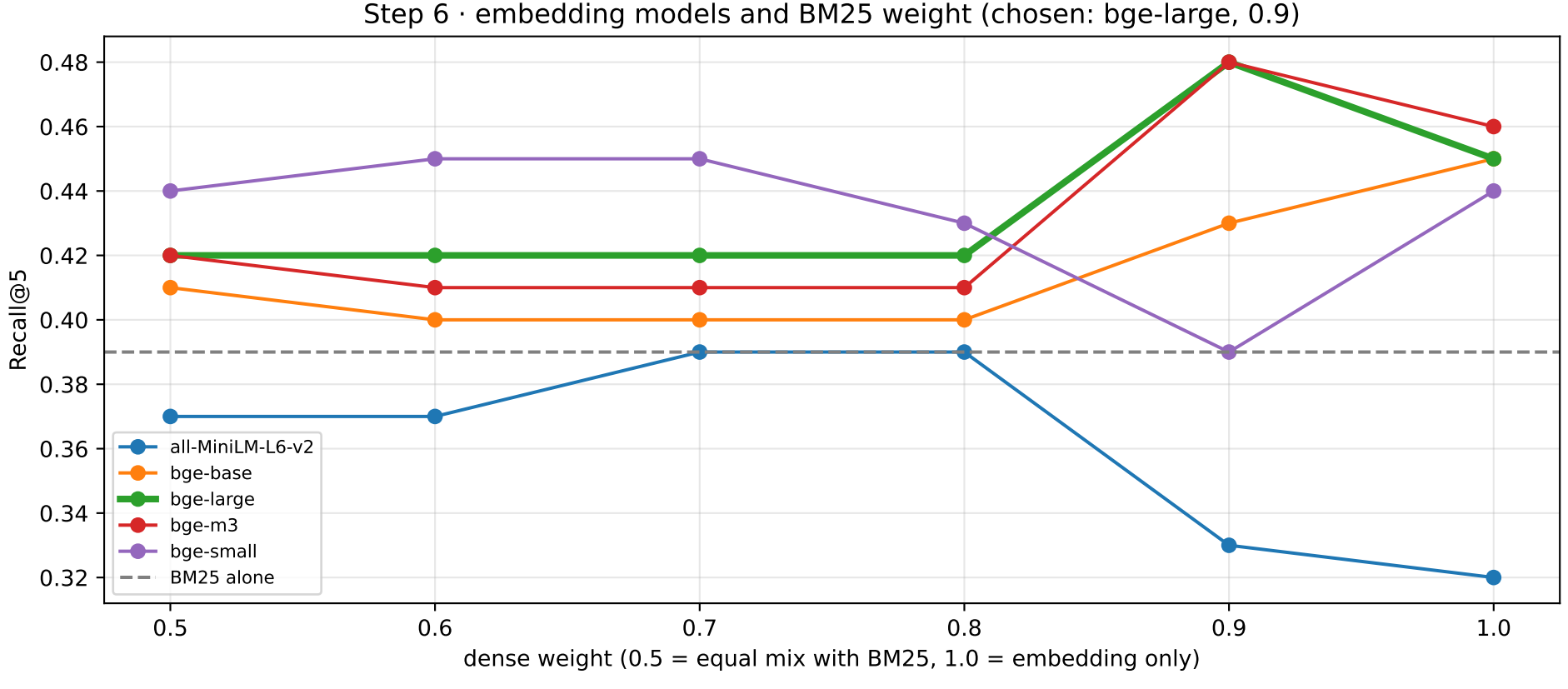
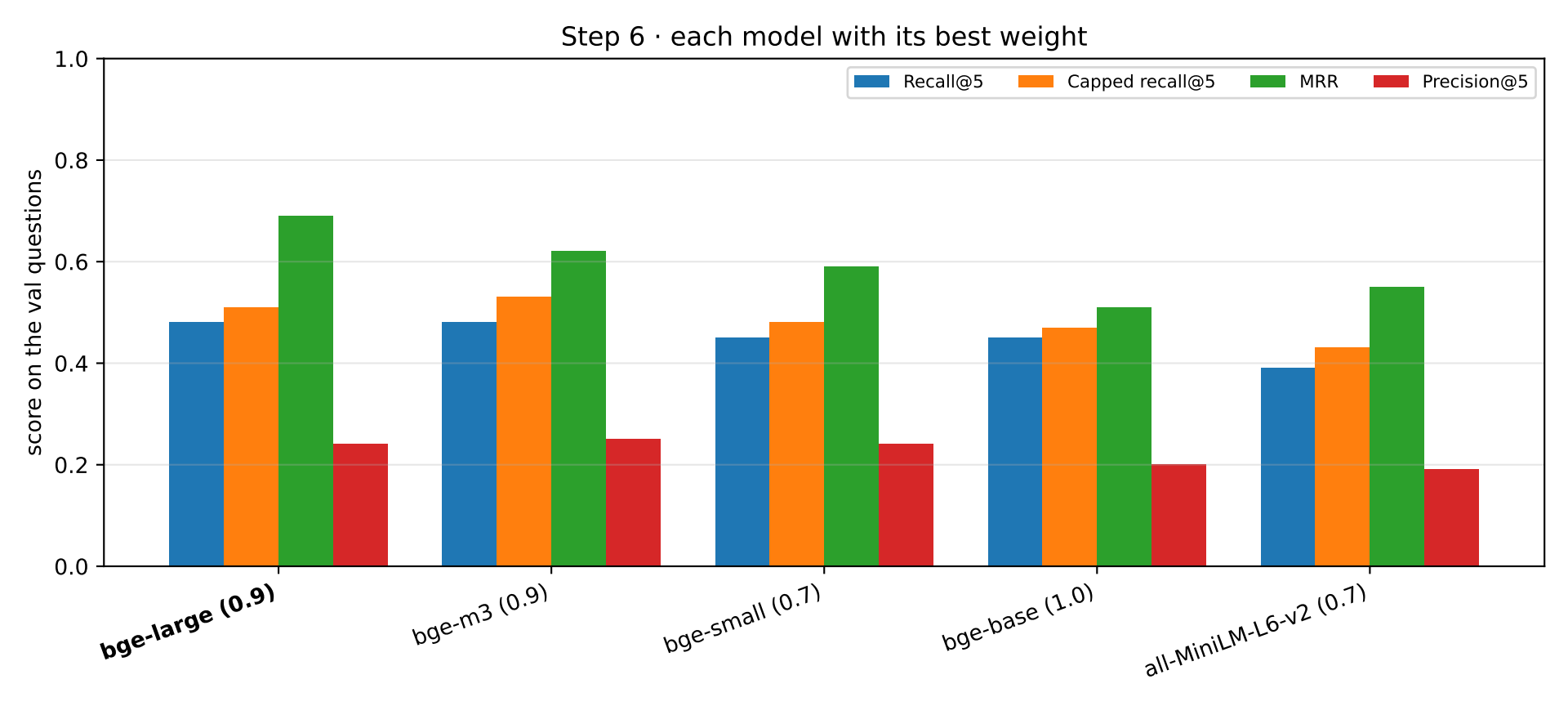
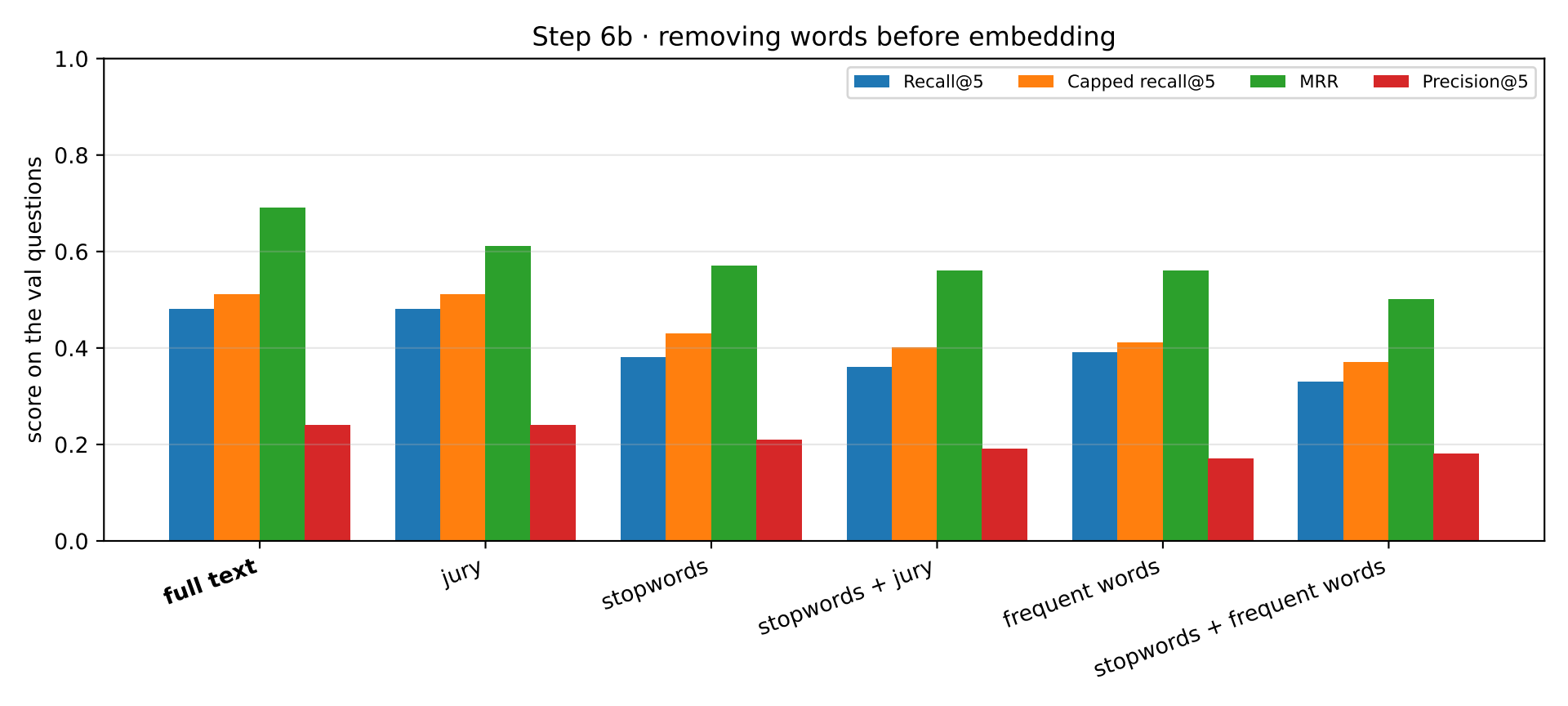
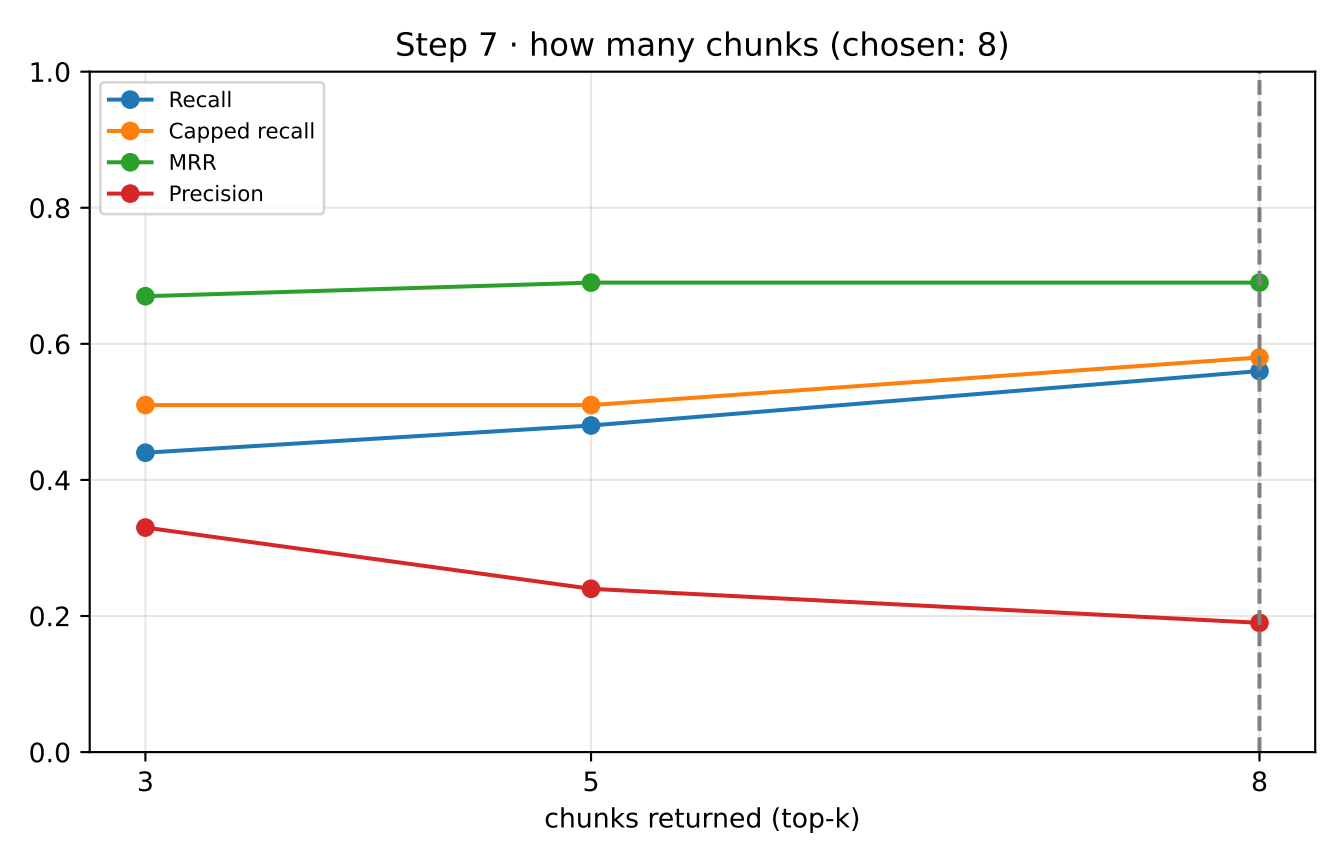
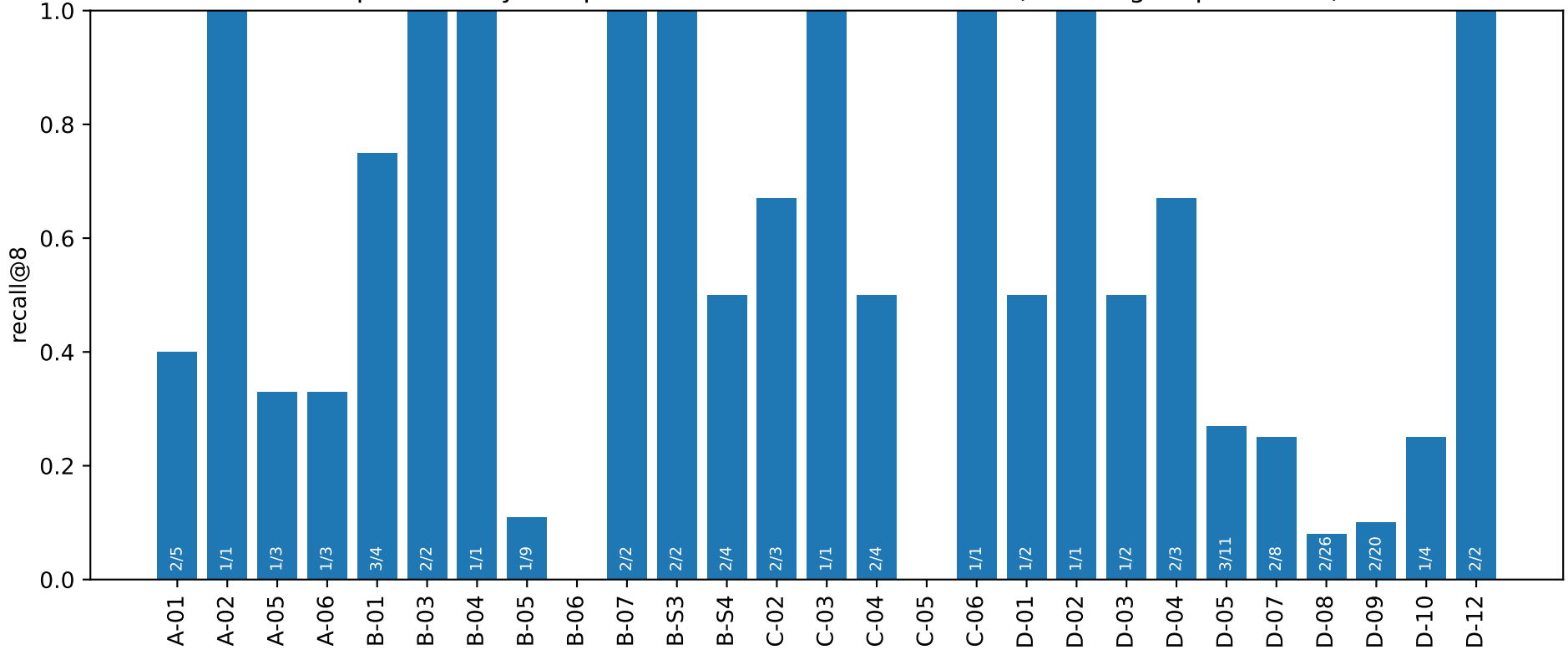

Step 5 · BM25: which words to remove
Recall@5
Capped recall@5
MRR
Precision@5
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
none
jury
stopwords
stopwords + jury
frequent words
stopwords + frequent words


Step 6 · embedding models and BM25 weight (chosen: bge-large, 0.9)
Recall@5
0.48
0.46
0.44
0.42
0.40
0.38
0.36
0.34
0.32
0.5
0.6
0.7
0.8
0.9
1.0
dense weight (0.5 = equal mix with BM25, 1.0 = embedding only)
all-MiniLM-L6-v2
bge-base
bge-large
bge-m3
bge-small
BM25 alone


Step 6 · each model with its best weight
Recall@5
Capped recall@5
MRR
Precision@5
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
bge-large (0.9)
bge-m3 (0.9)
bge-small (0.7)
bge-base (1.0)
all-MiniLM-L6-v2 (0.7)


Step 6b · removing words before embedding
Recall@5
Capped recall@5
MRR
Precision@5
score on the val questions
1.0
0.8
0.6
0.4
0.2
0.0
full text
jury
stopwords
stopwords + jury
frequent words
stopwords + frequent words


Step 7 · how many chunks (chosen: 8)
Recall
Capped recall
MRR
Precision
1.0
0.8
0.6
0.4
0.2
0.0
3
5
8
chunks returned (top-k)


Step 10 · every val question with the chosen search (found / gold provisions)
recall@8
1.0
0.8
0.6
0.4
0.2
0.0
2/5
1/1
1/3
1/3
3/4
2/2
1/1
1/9
2/2
2/2
2/4
2/3
1/1
2/4
1/1
1/2
1/1
1/2
2/3
3/11
2/8
2/26
2/20
1/4
2/2
A-01
A-02
A-05
A-06
B-01
B-03
B-04
B-05
B-06
B-07
B-S3
B-S4
C-02
C-03
C-04
C-05
C-06
D-01
D-02
D-03
D-04
D-05
D-07
D-08
D-09
D-10
D-12